Jawher Jabri

AI Research Engineer

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Summary

AI Research Engineer with 3+ years of experience in deep reinforcement learning and LLM systems. Designs rewards, action spaces and training environments for industrial scheduling and optimization, and runs controlled experiments against strong baselines to show what works. Also builds, evaluates and deploys LLM systems and agent tooling in production.

Professional Experience

AI Research Engineer, InstaDeep Ltd, Tunis, Tunisia Oct 2023 to Present

- **Reinforcement Learning for Train Scheduling, Deutsche Bahn**
  - Work on the design of the agent's action space and of the train simulation environment it is trained in.
  - Redesigned the RL agent's action space using shortest-path search on the rail network, staying closer to the original timetable and cutting train delays by **30%** compared with the previous agent.
  - Profiled the train simulation environment and cached expensive computations in its slowest code paths, raising simulation throughput by **60%**.
- **Reinforcement Learning for PCB Placement, DeepPCB**
  - Trained and debugged Proximal Policy Optimization (PPO) agents in JAX, with experiments configured in Hydra and tracked in MLflow.
  - Designed a new placement sub-reward that follows how PCB designers work, which led the routing agent to use **10%** fewer vias.
  - Developed a Linformer (linear-attention) model from scratch in JAX to replace the quadratic memory cost of standard attention, growing the largest board that fits on a 40 GB GPU from about 300 to about **500** components.
  - Created a probabilistic, geometry-based placement heuristic in JAX, parallelized for speed, improving runtime by **30%** and solution quality by **20%**.
- **LLM Extraction Pipeline and Evaluation Harness, DeepPCB**
  - Built an end-to-end pipeline that extracts structured data from PDF documents with Gemini on Vertex AI and the OpenAI Assistants API, reaching **83%** correct answers on the evaluation set.
  - Added a heuristic extraction from the PCB board files and combined it with the PDF results, raising accuracy to **89%**.
  - Processed each PCB board's PDFs in parallel and merged the results, cutting extraction time per PCB board from about 3 minutes to about **30 seconds**.
  - Set up a fixed evaluation set scored against ground truth and ran it as pytest regression tests, so every prompt, schema or model change is measured against the same reference.
  - Measured retrieval and generation separately, to tell whether an error came from pulling the wrong part of the document or from the model misreading the right part.
  - Compared Gemini and OpenAI on accuracy, latency and cost, and chose Gemini for production, which scored higher at a lower cost.
- **MCP Server for DeepPCB**
  - Exposed **9** DeepPCB product features as MCP tools, used by DeepPCB customers and by the AI agent integrated into the web app.
  - Deployed the MCP server to production on Ray Serve as a scalable service.
  - Worked with product and engineering stakeholders to turn research prototypes into supported product features.

Research Engineer Intern, Graduation Internship, InstaDeep Ltd, Tunis, Tunisia Apr 2023 to Sep 2023

- **FPGA Routing: Benchmark Generator, Fixed Baseline and RL Agents**
  - Wrote a benchmark generator that creates FPGA routing problems at chosen difficulty levels, so classical and RL solvers are compared on exactly the same problems.
  - Implemented a graph-based baseline router (pathfinding and combinatorial optimization over the interconnect graph) and used it as the reference for every later comparison.
  - Designed the RL environment and developed an RL agent that improves the baseline solver's solutions.

Data Science NLP Intern, Safran Tunisia, Soliman, Tunisia Jul 2022 to Aug 2022

- **Question Answering over Technical Documents**
  - Cleaned and parsed technical PDFs with inconsistent layouts into a searchable document collection.
  - Developed the retrieval and reader pipeline with Haystack and fine-tuned *roberta-base-squad2* with HuggingFace Transformers to improve answer accuracy.
  - Served the model through a FastAPI REST API so other applications could consume it.

Open-Source Projects

chronobox and chronobox-ops, Personal Open-Source Project Oct 2026 to Present

- **chronobox: Exact Undo and Forks of a Live Machine for Coding Agents**
  - Built an MCP server that gives coding agents (Claude Code and other MCP clients) their own Linux machine per session, a Firecracker microVM that boots in about 1 second.
  - Implemented checkpoint and rollback of the whole machine, including disk, memory, running processes and open database transactions, restored in about **50 ms**.
  - Implemented forking of the live machine into exact copies, so the agent tries a risky change on a copy while the real machine keeps serving users, then applies the verified change.
  - Added a rehearsal guard that refuses destructive commands on the real machine until they have succeeded on a fork, and a hand-off mode that replays the agent's script on a fresh copy under traffic, killed halfway and rerun.
- **chronobox-ops: Benchmark of Coding Agents on Live Production Servers**
  - Designed **37** tasks, each a real Linux machine with real services, data and clients writing during the run, plus a hidden trap that a careful engineer would avoid.
  - Built the grader: hidden probes record failed requests, lost writes and changed data every 3 seconds, and after the run the grader fast-forwards delayed effects (nightly jobs, counters near their limit) before checking that the job stays done.
  - Ran and published **1,477** runs of five Claude models, alone and with chronobox, with confidence intervals; every results table rebuilds from the published runs with one command.
  - Four of five models succeeded more often and harmed users less with chronobox. Sonnet 5.5 with chronobox matched Opus 4.8 working alone at **8 times** lower cost and halved its own harm rate, from 26% to 12%.
  - Reported where it does not help: the strongest model, Opus 5.5, already succeeds on 96% of runs alone and gains nothing.

Skills

Programming: Python

Data & Databases: SQL, PostgreSQL, NumPy, Pandas

Reinforcement Learning: Deep Reinforcement Learning, PPO, Policy Optimization, Reward Design, Action Space Design, Environment Engineering, Simulation, Distributed Training

RL Theory (taught): Actor-Critic, Deep Q-Networks (DQN), RLHF, GRPO

LLMs & Agents: Large Language Models (LLMs), Generative AI, RAG, MCP (Model Context Protocol), Function Calling, Structured Outputs, Prompt Engineering, Agent Evaluation, OpenAI API, Gemini

Deep Learning: Machine Learning, Deep Learning, Transformer Models, Fine-tuning

Frameworks: PyTorch, JAX, TensorFlow/Keras, Scikit-learn, HuggingFace Transformers, Haystack, Ray, RLlib, Gymnasium

Experimentation: Experiment Design, Controlled Comparisons, Benchmark Design, Model Evaluation, Failure Attribution, Regression Testing

Experiment Tracking: Hydra, MLflow, Weights & Biases

Graphs & Optimization: Graph Modeling, Shortest-Path Search, Pathfinding, Heuristic Solvers, Combinatorial Optimization

Engineering: Model Deployment, Docker, FastAPI, Ray Serve, REST APIs, Firecracker, Linux, Unit Testing, Debugging, CI/CD, Git, Google Cloud Platform (GCP, Vertex AI)

Education

MSc in Big Data, Artificial Intelligence, Paris Dauphine University PSL, Sep 2021 to Oct 2023

Bachelor in Business Computing, Higher Institute of Management Tunis, Sep 2018 to Jun 2021

Languages

English (Proficient), French (Intermediate), Arabic (Native)